IREM B. GÜNDÜZ

PhD Candidate · Computational Epigenomics & Machine Learning

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▪ PROFILE

Machine-learning researcher in genomics, focused on models that are accurate and interpretable in equal measure. I design graph neural networks that embed real biological structure, such as transcription-factor interaction networks, directly into how they learn, in order to study how cells establish and change their identity. My work spans single-cell and bulk multi-omics, explainable AI, and causal inference, and has contributed to studies of the human immune epigenome (Nature Genetics, 2025). I also develop and maintain open-source genomics software in R and Python, turning methods into tools the community can use.

▪ TECHNICAL SKILLS

Programming: R (expert; Bioconductor / package development), Python (machine learning & deep learning), Bash

ML / Deep Learning: PyTorch, PyTorch Geometric (GNNs, GATv2), scikit-learn; graph neural networks, explainable AI, causal inference, attention-based models

Genomics: single-cell & bulk multi-omics — scATAC-seq, snmC-seq2, RNA-seq, ATAC-seq, DNA methylation; TF-activity inference, deconvolution, regulatory-network analysis

Tools: Git / GitHub, conda, high-performance computing, reproducible workflows

▪ PUBLICATIONS & PREPRINTS

Peer-reviewed

Wang, **Gündüz IB**, et al. Genetics and environment distinctively shape the human immune cell epigenome. *Nature Genetics*, 2025. *(second author)*

John A, **Gündüz IB**, et al. LLMRS: Unlocking Potentials of LLM-Based Recommender Systems for Software Purchase. *33rd Workshop on Information Technologies and Systems (WITS) 2024*, accepted. *(second author; with Prof. W. Maass, DFKI)*

Cuadrat RRC, **Gündüz IB**, et al. Cardiovascular disease biomarkers derived from circulating cell-free DNA methylation. *NAR Genomics and Bioinformatics*, 2023. *(second author; ML-based deconvolution, deconvR)*

Baranovskii A*, **Gündüz IB***, Franke V, Uyar B, Akalin A. Multi-Omics Alleviates the Limitations of Panel Sequencing for Cancer Drug Response Prediction. *Cancers*, 2022; 14(22):5604. *(*joint first author)*

Preprints / under review

Gündüz IB, et al. Dissecting the epigenome dynamics in human immune cells upon viral and chemical exposure by multimodal single-cell profiling. Preprint, 2025. *(first author; primary PhD project)*

Manuscripts in preparation (first author)

- **methylTFR** — quantifying transcription-factor activity from DNA-methylation signatures at TF-binding sites; recovers lineage-specific TFs across human immune cell types (R package).
- **ChrAccR** — a unified, memory-efficient framework for single-cell and bulk chromatin-accessibility (ATAC-seq) analysis (R package).
- **ARTEMIS** — attention-based skip-connection GNN (GATv2) embedding TF–TF interaction priors for interpretable cell-type prediction; sole developer (Python, PyTorch Geometric).
- **Multi-modal characterization of CD8 T-cell exhaustion** across acute, chronic, and tumoral stages, integrating single-cell multiome atlases across viral cohorts (co-supervised MSc thesis).

▪ SOFTWARE

- **deconvR** — bulk-sample deconvolution from reference signature profiles with multiple modeling options; applied in a cardiovascular cfDNA-methylation biomarker study (NAR Genomics & Bioinformatics, 2023). Bioconductor (developer & maintainer).
- **methylTFR** — TF-activity analysis from methylation signatures. github.com/EpigenomeInformatics/methylTFR
- **ChrAccR** — unified chromatin-accessibility analysis. github.com/EpigenomeInformatics/ChrAccR
- **ARTEMIS** — interpretable GNN for regulatory TF explanation from methylation signatures (in preparation). github.com/EpigenomeInformatics/ARTEMIS

▪ RESEARCH EXPERIENCE

PhD Researcher — ICBB Lab **May 2023 – present**

Integrative Cellular Biology & Bioinformatics, Saarland University, Saarbrücken · Supervisor: Jun.-Prof. Dr. Fabian Müller · (expected completion 2027)

- Developed ARTEMIS, a sole-authored attention-based GNN that embeds TF–TF interaction priors for interpretable cell-type prediction (Python, PyTorch Geometric).
- First-authored a multimodal single-cell study of immune-cell epigenome dynamics under viral and chemical exposure (primary PhD project).
- Authored and maintain methylTFR and ChrAccR for TF-activity and chromatin-accessibility analysis.
- Co-supervise an MSc thesis on multi-modal CD8 T-cell exhaustion; Teaching Assistant for Single-Cell Bioinformatics and related courses.

Preparatory-Phase PhD Researcher **Dec 2021 – May 2023**
Saarland University Graduate School of Computer Science, Saarbrücken

- Single-cell epigenomic analysis of pathogen-exposed individuals using statistical-learning approaches (ICBB Lab, F. Müller).
- Applied machine-learning models to quantify alternative-splicing events and analyze transcriptional-network breakdown in aging/parabiosis (Chair of Clinical Bioinformatics, CBI; A. Keller).

Bioinformatics Intern — Bioinformatics & Omics Data Science **Jul 2021 – Jun 2022**
Berlin Institute for Medical Systems Biology (BIMSB), MDC Berlin · Supervisor: Dr. Altuna Akalin

- Contributed to peer-reviewed ML work on cancer drug-response prediction (Cancers, 2022) and cell-free-DNA-methylation deconvolution (NAR Genomics & Bioinformatics, 2023).
- Developed the deconvR package (Bioconductor) for reference-based sample deconvolution.

Project Intern — Bishop Laboratory **Apr 2021 – Aug 2021**
Bioinformatics Research Network · Supervisors: Dr. Alexander J. R. Bishop, Dr. Daniel Montemayor

- Built ensemble ML models using R-loop levels to predict cancer tissue status in clinical settings.

Student Assistant — Computational Biology & Bioinformatics Lab **Oct 2019 – Aug 2021**
Marmara University, Istanbul · Supervisor: Assoc. Prof. Dr. Pemra Özbek Sarıca

- TÜBİTAK 1001-funded computational drug discovery targeting N. meningitidis inhibition.

Earlier: *wet-lab research internships — Vem Pharmaceuticals (drug-production QC) and Aziz Sancar Institute of Experimental Medicine, neuroscience lab (western blot, ELISA), 2020.*

EDUCATION

Ph.D., Computer Science — Saarland University **2023 – present (expected 2027)**
Faculty of Mathematics & Informatics · cumulative thesis

Fast-track / Preparatory Phase — Saarland University **2021 – 2023**
Graduate School of Computer Science

B.Eng., Bioengineering — Marmara University **2017 – 2021**
Faculty of Engineering, Istanbul · Honor Student

TEACHING & MENTORING

- Teaching Assistant, Single-Cell Bioinformatics *(every winter semester)* — Saarland University.
- Co-supervisor, MSc thesis (multi-modal CD8 T-cell exhaustion).
- Teaching Assistant, CompGen 2021: Hands-on Course on Machine Learning for Genomics — prepared benchmark ML models and mentored capstone projects.

AWARDS & SCHOLARSHIPS

- Open Science Reporter — Saarland University (2022).
- Preparatory-Phase PhD Scholarship — Saarland University Graduate School of Computer Science (2021–2023).
- TÜBİTAK 1001 Research Scholarship — Scientific & Technological Research Council of Turkey (2020–2021).
- Honor Student — Marmara University, Department of Bioengineering (2021).

SELECTED TALKS & ACTIVITIES

- Poster, RECOMB 2023 — Chromatin accessibility landscape of human peripheral blood cells upon pathogen exposure.
- UAI 2022 (Uncertainty in Artificial Intelligence); e:Med Summer School 2022 — single-cell-based systems medicine.
- SCOG Workshops — Recent Advances in Single-Cell Epigenomics (2022, 2023, 2024).

LANGUAGES

Turkish (native) · **English** (fluent) · **German** (intermediate)